

5. Limites trigonométricos

$$1.) \lim_{x \rightarrow 0} \frac{\operatorname{sen} \frac{x}{2}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\operatorname{sen} \frac{x}{2}}{\frac{x}{2}} \cdot \frac{1}{2}$$

$$= 1 \cdot \frac{1}{2}$$

$$= \frac{1}{2} //$$

$$2.) \lim_{\theta \rightarrow 0} \frac{\operatorname{sen}^2 \frac{\theta}{3}}{\theta^2}$$

$$= \lim_{\theta \rightarrow 0} \left(\frac{\operatorname{sen} \frac{\theta}{3}}{\theta} \right)^2$$

$$= \left(\frac{1}{3} \right)^2$$

$$= \frac{1}{9} //$$

$$3.) \lim_{w \rightarrow 0} \frac{-4(\cos w + 4)}{8w}$$

$$= \lim_{w \rightarrow 0} \frac{4(1 - \cos w)}{8w}$$

$$= \lim_{w \rightarrow 0} \frac{1 - \cos w}{2w}$$

$$= 1 - \cos w = 0$$

$$= 0 //$$

$$4.) \lim_{\theta \rightarrow 0} \frac{\theta}{\operatorname{sen} \theta}$$

$$= \frac{1}{\frac{\operatorname{sen} \theta}{\theta}}$$

$$= \frac{1}{1}$$

$$= 1 //$$

$$5.) \lim_{\theta \rightarrow 0} \frac{\sec 2\theta \tan 3\theta}{5\theta}$$

$$= \frac{1 \cdot 3}{5}$$

$$= \frac{3}{5} //$$

$$6.) \lim_{\theta \rightarrow 0} \frac{2 \tan^2 \theta}{\theta^2}$$

$$= 2 \lim_{\theta \rightarrow 0} \left(\frac{\tan \theta}{\theta} \right)^2$$

$$= 2(1)^2$$

$$= 2 //$$

$$7.) \lim_{t \rightarrow 0} \frac{3t^2}{1 - \cos^2 \frac{t}{2}}$$

$$= 1 - \cos^2 x = \sin^2 x$$

$$= \lim_{t \rightarrow 0} \frac{3t^2}{\sin^2 \frac{t}{2}}$$

$$= 3 \left(\frac{t}{\sin \frac{t}{2}} \right)^2$$

$$= 3(2)^2$$

$$= 12 //$$

$$8.) \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{2x^2}$$

$$= 1 - \cos^2 x = \sin^2 x$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{2x^2}$$

$$= \frac{1}{2} \left(\frac{\sin x}{x} \right)^2$$

$$= \frac{1}{2} (1)^2$$

$$= \frac{1}{2} //$$

$$9.) \lim_{\theta \rightarrow 0} \frac{\theta^2 + 3\theta}{\sin \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{\theta(\theta + 3)}{\sin \theta}$$

$$= \lim_{\theta \rightarrow 0} (\theta + 3) \frac{\theta}{\sin \theta}$$

$$= (0 + 3)(1)$$

$$= 3 //$$